

Sourabh Singha

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Course	Displicine	College/University	Year	CGPA/10
Ph.D.	Chemical Engineering	IIT Bombay	2021-2026	9.21
M.Tech	Chemical Engineering	IISc Bangalore	2018-2020	7.40 (Department Topper)
B.E	Chemical Engineering	Jadavpur University	2014-2018	7.37

- **Objective:** Seeking a role in the energy industry—particularly in hydrogen storage materials—where I can apply my technical expertise in materials science and molecular dynamics, contribute to innovative solutions, and further develop my skills for mutual growth and advancement.

ACADEMICS DETAILS

- **Ph.D. Thesis: Hydrogen Storage Materials based on Thermodynamics and Kinetics approach** [2021-2026]
  - Targeting a hydrogen storage system with gravimetric capacity exceeding 10 wt.% to enable high energy density storage. Aim to achieve over 1500 charge-discharge cycles with hydrogen release below 353 K (80°C) for practical and safe operation.
  - Critical performance metrics include rapid hydrogen adsorption/desorption kinetics, favorable thermodynamics characterized by optimal enthalpy and entropy values, and efficient, reversible hydrogen release mechanisms.
  - Addressing challenges across different hydrogen storage states, with a particular focus on solid-state storage due to its superior scalability, enhanced safety profile, reduced risk of leakage or explosion, and simplified handling and transportation logistics for large-scale applications.
  - Employ computational modeling to optimize material properties and predict hydrogen storage behavior, accelerating the design and development of advanced hydrogen storage materials with high cycle stability and energy efficiency.

  Reference: *Prof. Abhijit Chatterjee, IIT Bombay, India, abhijit@che.iitb.ac.in*
- **M.Tech Thesis: Soluble lead Redox Flow Battery based on Bipolar Plate** [2018-2020]
  - My work involves the study of the use of a bipolar plate in a soluble lead Redox flow battery. A soluble lead redox flow battery is a potential rechargeable battery. This system has the capacity to store energy at a large scale which will be very helpful to the renewable sector. We have observed an increase in voltage to around 4.2 V for a single cell.
  - We believe that this technology has the potential to replace the existing other batteries in certain applications of the electric device like mobile, power banks, and electric vehicles. I performed an experiment as well as validating the results using a simulation (COMSOL). Redox flow battery has advantages such as high potential, high efficiency, better performance, long cycle life, simple design, and low cost.

  Reference: *Prof. Sanjeev Kumar Gupta, IISc Bangalore, India, sanjeev@iisc.ac.in*
- **B.E Thesis: Catalytic Reduction of Carbon dioxide/Carbon monoxide to methanol** [2014-2018]
  - CO_2 produces highly volatile organic compounds (VOC) and is bad for the environment. The objective is to convert CO_2 to methanol. Different types of reactors were used such as PFR, CSTR, packed bed reactor, and auto-catalytic reactor, and finally, the focus was on which reactor performed better and provided high efficiency.
  - Methanol can be used as an alternative biofuel for internal combustion and other engines. It is also used to make other chemicals such as formaldehyde. I have done the project experimentally and theoretically.

  Reference: *Prof. Sudeshna Saha, Jadavpur University, India, saha.sudeshna@gmail.com*

WORK EXPERIENCE

- **Post Graduate Engineering Trainee | Tata Advanced System Limited, Hyderabad, India** [Aug'2020-Feb'2021]
  - Special processes such as anodizing, corrosion resistance, chemical conversion, plating, primer, topcoat, and coating. The main objective was to convert trivalent chromium from hexavalent chromium. Initially, I had performed it theoretically as well as experimentally.

- Performing theoretically as well as experimentally. Working on various projects: Sikorsky -S92 Helicopter, Lockheed Martin, Boeing, Cobham mission systems, Rolls Royce, GE Leap Engines, Airbus.

- **Summer Internship | Waterborne Coatings for OEM BASF, MANGALORE** [May'2019-July'2019]
  - The main objective of the project was to identify low-temperature curable waterborne coating. To achieve the goal of the project used an amine-blocked acid catalyst to cure standard waterborne coatings at lower temperatures. There were different amines used based on their PKa values varying from 4-12. This study used 3C1B system for curing and baking was conducted at 88°C, 66°C, and 50°C respectively.
  - The best thing was that most of the formulations were cured at 66°C and pot life was more than 168 hours. The cured coatings were tested for solvent resistance, hardness, and adhesion.

SKILLS & INTERESTS

- **Programming Languages**: FORTRAN90, Python, Matlab, C/C+, R, Bash
- **Tools & Libraries**: LAMMPS, VASP, DFT, VMD, OVITO, SQL, Material Studio, VESTA, Quantum expresso, COMSOL, CFD, Latex, Linux
- **Interests**: Automatic Code, Mathematical Modeling, Kinetics Modeling, Data Analysis, Machine Learning

HONORS and CERTIFICATE

- **Honors**
  - OASIS Scholarship [Jun'2014-Jun'2018]
  - C R Dasgupta Scholarship [2015]
  - MHRD Scholarship [Aug'2018-July'2020]
  - TAP Scholarship [Jan'2022-Dec'2026]
- **Certificates**
  - MATLAB Certificate Course
  - Certification in Madhava Mathematics Competition
  - RSS volunteer at IIT Bombay

RELEVANT COURSES

- **Molecular Simulation | Advanced Thermodynamics | Mathematical Methods | Advanced Reaction Engineering | Optimization** [Ph.D.]

- **Advanced Mathematics | Chemical Reaction Modelling | Molecular Dynamics | Numerical Method | Interfacial and colloidal |Statistical Thermodynamics** [M.Tech]

- **Chemical Reaction Engineering |Heat transfer/mass transfer | Thermodynamics |Chemical Techonology | Fluid Mechanics** [B.E]

PUBLICATION & CONFERENCE

- Singha, Sourabh, and Abhijit Chatterjee. "Atomistic-scale insights into hydrogen diffusion barrier in nickel hydride: Complex interplay between short-and long-range hydrogen arrangement and hydrogen concentration." Computational Materials Science 241 (2024): 113044.
- Singha, Sourabh, and Abhijit Chatterjee. "Computational workflow for investigating hydrogen permeation in novel hydrogen storage materials." arXiv e-prints (2023): arXiv-2312.
- Singha, Sourabh, and Abhijit Chatterjee, A chemo-thermo-mechanical model for predicting size-dependent enthalpy and volume changes during formation of nickel hydride (to be submitted).

WORKSHOP ATTENDED

- AI IMPACT 2023 (Fostering Collaborations for Impactful AI Technologies) (attended, April 28-29, 2023)